

EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 32-300-1

DATE: 10-10-2023

Supra-Harmonic Power Quality Analyzer

EDMS 32-300 -1
SPECIFICATION
FOR
Supra-Harmonic Power Quality Analyzer

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EEHC DISTRIBUTION MATERIALS SPECIFICATION**EDMS 32-300-1****DATE: 10-10-2023****Supra-Harmonic Power Quality Analyzer****CONTENTS**

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EEHC DISTRIBUTION MATERIALS SPECIFICATION**EDMS 32-300-1****DATE: 10-10-2023****Supra-Harmonic Power Quality Analyzer****Abbreviations**

AC: Alternating Current

IEC: International Electro Technical Commission

IEEE: Institute of Electrical and Electronics Engineers

Pst: Short Term Flicker Perceptibility

Plt: Long Term Flicker Perceptibility

RMS: root mean square

TFT: thin Film transistor

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This specification is for a power Quality Analyzer (further herein referred simply as "the analyzer". The analyzer shall have the capability of examining and recording the true condition of any low voltage electrical system . It shall provide complete and accurate True RMS measurements of all important electrical parameters for single and three phase application. The analyzer shall be portable

2. APPLICABLE STANDARDS

Unless otherwise specified in this specification, the offered analyzer shall comply with all the latest relevant applicable standards Issued by "International Electro Technical Commission (IEC)"

S.No	Standard No.	Description
1	IEC 61000-4-1	Electromagnetic Compatibility (EMC) –Part 4-1 Testing and measurement techniques- Overview of IEC 61000-4 series
2	IEC 61000-4-7	Electromagnetic Compatibility (EMC)- Part 4,7: Testing and measurement techniques- General guide on harmonics and inter harmonics measurements and instrumentation , for power supply systems and equipment connected thereto
3	IEC 61000-4-30	Power Quality Measurement methods
4	IEC 61000-4-15	Testing and measurement techniques- Flicker meter- Functional and design specifications
5	IEC 61010-1	Safety requirements for electrical equipment for measurement, control and laboratory use - part 1: General.
6	IEC 61000-2-2	Environment - Compatibility levels for low-frequency conducted disturbances and signaling in public low-voltage power supply systems
7	IEC 61000-2-4	Electromagnetic compatibility (EMC) - Part 2-4: Environment - Compatibility levels in industrial plants for low-frequency conducted disturbances
8	IEC 61000-3-2	Electromagnetic compatibility (EMC) - Part 3-2: Limits - Limits for harmonic current emissions (equipment input current: ≤ 16 A per phase)
9	IEC 61000-3-12	Electromagnetic Compatibility (EMC)-Part 3-12 Limits –Limits for harmonic Current Produced by equipment connected to Low voltage system with input current >16 A and ≤ 75 A per phase
10	IEC 61000-4-5	Electromagnetic compatibility (EMC) - Part 4-5: Testing and measurement techniques - Surge immunity test
11	IEEE 519	IEEE Recommended Practice and Requirements for Harmonic Control in Electric Power Systems

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The performance of analyzer should be guaranteed for following environmental conditions, any differences in the guaranteed performance should be clearly set out in the offer.

Ambient air temperature	-5°C to 45 °C (50°C as option)
Maximum relative humidity	90%
Altitude	≤1000m

The analyzer shall be used for measurement of various electrical parameters in domestic, industrial, commercial meter installation, and distribution network circuit.

4. DESIGN AND CONSTRUCTION

- 4.1. The analyzer measurement shall be according to IEC 61000-4-30 class A
- 4.2. The analyzer shall accurately measure Voltage (V), current (A), Power factor (%), lag and lead), true power (kW), reactive power (kVAR) and apparent power (kVA) along with voltage and current harmonics in electrical systems
- 4.3. The analyzer shall have 4 channel voltage to measure True phase- phase as well as phase to neutral voltage, with Minimum and Maximum registers, and Voltage Unbalance (+ve sequence, -ve sequence, zero sequence).
- 4.4. The analyzer shall have measure True phase- phase as well as phase to neutral voltage represented as phasor and values
- 4.5. The analyzer has the capability of measuring Voltage Disturbance parameters such as (transients, dips, swell, flicker (Plt, Pst)) at the same time ,the measurements data shall be presenting in graph and values
- 4.6. The analyzer shall have 4 channel to measure True RMS current in each phase, with Minimum and, Maximum registers, phase displacement, neutral current and current Unbalance (+ve sequence, -ve sequence, zero sequence).
- 4.7. The analyzer shall measure True RMS as well as displaced power factor of each phase with indication of Minimum and Maximum values of measurements
- 4.8. The analyzer shall indicate Angle Displacement between voltage and current, as well as the three phase voltage sequence.
- 4.9. The analyzer shall measure Active, Reactive and apparent power per phase and also total values for three phases.
- 4.10. The analyzer shall measure the energy power kwh and kvarh during the recording period
- 4.11. The analyzer shall measure voltage and current Harmonics (individual and inter-harmonic) up to 50th order for each phase expressed as a percentage of the fundamental or as value.

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- 4.12. The analyzer shall measure the total harmonic distortion of voltage and current and has the capability for calculating the Total demand distortion for the measured period as defined in IEEE 519.
- 4.13. The analyzer shall measure the supra-harmonic for voltage in the range of (2-150) kHz.
- 4.14. It is preferable that the analyzer should have the ability to measure the supra-harmonic above (2.5 kHz) current.
- 4.15. The analyzer shall have probes for easy connection for both single phase and three phase application.
- 4.16. The analyzer shall have color (thin film transistor) TFT display at least 4.5 inch to give actual measured values and waveform for electrical parameters such as: voltage, current, power factor, phase angle for each phase at site.
- 4.17. The portable analyzer shall be light in weight not more than 3Kg.
- 4.18. The analyzer's degree of protection shall be IP65.
- 4.19. The analyzer shall continue recording in the same measurement file in case of power interruption.
- 4.20. The analyzer shall have the capability to be programmable at site through keypad without laptop.
- 4.21. The analyzer shall have secured application download to mobile to ease communication at site.

5. RATINGS

- 5.1. The analyzer shall be supplied by main voltage of 100 V to 230 V AC, 50 Hz supply protected by fuse.
- 5.2. The analyzer power consumption shall be low as possible
- 5.3. The analyzer has the capability of adjusting measuring interval at least 1sec values
- 5.4. The analyzer shall have rechargeable battery power option with the following characteristics:
- Lithium ion battery with life time at least 8 years and changeable
 - Shall have the ability to power the analyzer for at least 2 hours without main supply.
 - The analyzer battery shall have indication of charging percentage level.

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- 6.1. The analyzer shall have four flexible Rogowski coils rated at 3000A / 6000 A (according to EDC needs) with diameter not less than 19 cm.
- 6.2. The analyzer shall have four clamps up to 100 A.
- 6.3. The analyzer shall have four cables voltage clips crocodile type (3 phases, neutral) for single phase or three phase voltage measurement rated at 1000 VAC.
- 6.4. The analyzer shall have a USB port for data transfer to laptop computer.
- 6.5. The analyzer shall have software for data collection, analysis, representation (numerical & graph) and verification and recorded file will be downloaded in suitable file in compatible format to be opened in Microsoft excel file
- 6.6. The software shall be delivered in CD or USB flash memory for each analyzer.
- 6.7. Each analyzer shall be supplied with laptop computer (optional). The specification will be as in Annex1.
- 6.8. The analyzer software shall be compatible with Microsoft windows 10 operating system and all lower versions up to Microsoft Windows 7.
- 6.9. The analyzer shall have carrying case to fit the device and all its accessories and have suitable IP for protecting the equipment not less than 41.
- 6.10. The analyzer shall be designed for long term recording with memory capacity not less than 8 GB.

7. MARKING

The Analyzer shall be marked legibly and indelibly with the following information:

- Name or trade mark of the manufacturer
- Country of origin
- Type/model serial number
- Nominal input voltage and frequency
- Fuse ratings.

All marking shall to be written in English with suitable font.

8. TESTS

- A Copy of certificate of calibration on the Analyzer's performance across its operating range .The certificate for the type of analyzer being offered shall be from international calibration laboratory and provided with submitted tender documents.
- The manufacturer's declaration of conformity to reference standard shall be submitted.

9. SUBMITTILES

- Relevant technical detail, schematic drawing, operational and service manuals shall be submitted to support the tender and shall be clearly marked to indicate the type/model and serial number of analyzer being offered.
- The bidder shall submit all required certificate and the specification of Analyzer laptop

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- The analyzer shall have a warranty against any defects, which may develop due to faulty material, calibration, transportation for a period of two years from the date of delivery. All defective analyzers shall be replaced at supplier's cost.
- The supplier shall commit himself to make available essential spare parts and other consumables for a period not less than 5 years or for the expected life period of the analyzer whichever is greater.

11.TRAINING

The tenderer shall be committed to train at least 10 engineers on downloading software program on laptop, installing at location, and downloading the measurements and analysis with free charge.